# Validity of the halting implication entails that the machine halts

The Open Logic Project contributors

openlogic:turing-machines/undecidability:local-snapshot-20261013;
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*Editorial note.* The counted statement is the author's own lemma environment lem:halt-if-valid (source lines 1861-1864, the \ollabel on line 1862); the proof environment begins two lines below it and is its own proof, closing at source line 1894.

*Editorial note.* Neither the statement nor the proof contains an \olref or a \cite, so no source pointer and no bibliography entry are needed; every object used (T(M,w), E(M,w), the predicates Q_q and S_sigma, the structure M) is introduced by the author inside the retained unit and every symbol is declared in the retained preamble.

*Editorial note.* The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

*Editorial note.* Section and theorem counters are reset to zero so the retained lemma prints with the numbering of the retained front matter rather than continuing the frozen file's count.

*Editorial scope.* One original result of the Open Logic Project's undecidability unit, counted once: validity of the halting implication implies that the machine halts.

*Allowed standard prerequisites (editorial).*

- The sentences T(M,w) and E(M,w). T(M,w) is the author's sentence collecting the axioms that describe the initial configuration and the transition steps of M on w; E(M,w) is the existential sentence expressing that at some time and square the machine is in a state scanning a symbol whose transition is undefined.. The proof instantiates the validity of the implication between these two sentences at the structure it builds (source lines 1876-1890).
- The structure M with domain the natural numbers. The Lang L_M-structure M on the natural numbers interpreting 0 as zero, the prime function as successor, $<$ as less-than, and the predicates Q_q and S_sigma as the relations of being, at time n, in state q scanning square m, respectively containing symbol sigma at square m.. The structure is defined at source

lines 1867-1887 and is the only structure at which the assumed validity is instantiated.

**Lemma 0.1.** *If* $\models !T(M,w) \rightarrow !E(M,w)$*, then* $M$ *halts on input* $w$*.*

*Proof.* Consider the $\mathcal{L}L_M$-!!structure $\mathfrak{A}M$ with domain $\mathbb{N}$ which interprets 0 as 0, $'$ as the successor function, and $<$ as the less-than relation, and the predicates $Q_q$ and $S_\sigma$ as follows:

$$sQ_qM = \{\, \langle m,n\rangle \mid \begin{array}{l}\text{started on } w\text{, after } n \text{ steps,}\\ M \text{ is in state } q \text{ scanning square } m\end{array} \,\}$$

$$sS_\sigma M = \{\, \langle m,n\rangle \mid \begin{array}{l}\text{started on } w\text{, after } n \text{ steps,}\\ \text{square } m \text{ of } M \text{ contains symbol } \sigma\end{array} \,\}$$

In other words, we construct the !!structure $\mathfrak{A}M$ so that it describes what $M$ started on input $w$ actually does, step by step. Clearly, $M \models !T(M,w)$. If $\models !T(M,w) \rightarrow !E(M,w)$, then also $M \models !E(M,w)$, i.e.,

$$M \models \exists[x][\exists[y][(\bigvee_{\langle q,\sigma\rangle \in X} (Q_q(x,y) \wedge S_\sigma(x,y)))]].$$

As $|\mathfrak{A}|M = \mathbb{N}$, there must be $m$, $n \in \mathbb{N}$ so that $M \models Q_q(\underline{m},\underline{n}) \wedge S_\sigma(\underline{m},\underline{n})$ for some $q$ and $\sigma$ such that $\delta(q,\sigma)$ is undefined. By the definition of $\mathfrak{A}M$, this means that $M$ started on input $w$ after $n$ steps is in state $q$ and reading symbol $\sigma$, and the transition function is undefined, i.e., $M$ has halted. □